STAT201 · STATISTICS · LECTURE 2

Foundations of Probability Theory

Events, Kolmogorov's axioms, the addition and multiplication rules, independence and combinatorics

Syllabus topic 2 · week 2 · 4 academic hours

LECTURE GOALS

- write down a sample space and express events as its subsets;
- state the three Kolmogorov axioms and use them to reject impossible statements;
- apply the complement, addition and multiplication rules to business data;
- read joint, marginal and conditional probabilities from a contingency table and check independence;
- distinguish mutually exclusive events from independent events;
- count the number of ways using factorials, permutations, arrangements and combinations and apply them to compute probabilities.

1 From description to probability

Last week we described data that already exist: what happened. Now we turn to the question of what *may happen*. Probability is the language in which statistics speaks about uncertainty, and all the conclusions of the last weeks of the course depend on it: confidence intervals and p-values (weeks 9–13) are statements about probability.

Probability theory is built like geometry: first a few simple definitions and axioms, then the rules that follow from them. Once you master this lecture, you will be able to check any probability calculation for consistency.

2 Random experiment, sample space, events

BASIC CONCEPTS

Random experiment — a procedure whose result is not known in advance. **Sample space** S — the set of all possible results, each written once. **Event** A — any subset of S (a set of outcomes that we ask about).

EXAMPLE 1. ONE AND TWO DICE

One die: $S = \{1, 2, 3, 4, 5, 6\}$; the event "an even number came up" is the subset $A = \{2, 4, 6\}$. Two dice: S consists of 36 ordered pairs (i, j). The event "the sum equals 7" consists of six pairs: $(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)$.

1st \ 2nd	1	2	3	4	5	6
1	2	3	4	5	6	**7**
2	3	4	5	6	**7**	8
3	4	5	6	**7**	8	9
4	5	6	**7**	8	9	10
5	6	**7**	8	9	10	11
6	**7**	8	9	10	11	12

If all outcomes are equally likely, the probability of an event equals the share of favourable outcomes:

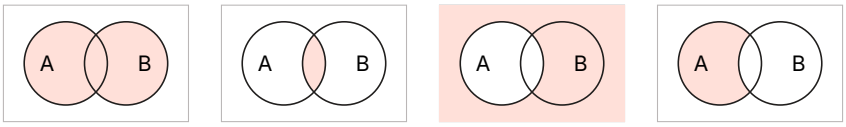
$$P(A) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}. \tag{2.1}$$

From the table: $P(\text{sum} = 7) = 6/36 = 0.167$; $P(\text{sum} = 12) = 1/36 = 0.028$; $P(\text{sum} > 9) = 6/36 = 0.167$; $P(\text{both even}) = 9/36 = 0.250$.

COMMON MISTAKE

The 36 ordered pairs are equally likely, but the 11 possible sums are not. A sum of 7 occurs six times as often as a sum of 12 (Fig. 1). Formula (2.1) may be applied only after checking that the outcomes really are equally likely.

2.1 Operations on events



$A \cup B$ — A or B (or both) | $A \cap B$ — A and B together | A' — not A | $A \cap B'$ — A but not B

- **Union** $A \cup B$ — A occurred, or B, or both.
- **Intersection** $A \cap B$ (also written AB) — both A and B occurred.
- **Complement** A' — everything that is not in A occurred.
- Events are **mutually exclusive** if they cannot occur at the same time: $A \cap B = \emptyset$. The decision on one loan application cannot be both "approved" and "rejected".

"Neither A nor B" is written as $(A \cup B)' = A' \cap B'$.

3 Kolmogorov's axioms and sources of probability

KOLMOGOROV'S AXIOMS, 1933

1. $0 \le P(A) \le 1$ for any event A.
2. $P(S) = 1$: a certain event has probability 1.
3. If A and B are mutually exclusive, then $P(A \cup B) = P(A) + P(B)$.

Everything else follows from these three axioms. For example, $P(\emptyset) = 0$, and if A is part of B, then $P(A) \le P(B)$. The main practical consequence: any statement about probabilities that violates the axioms is impossible. For example, one cannot have $P(A) = 0.20$ and $P(A \cap B) = 0.35$, because the intersection is part of A and cannot be more probable than it.

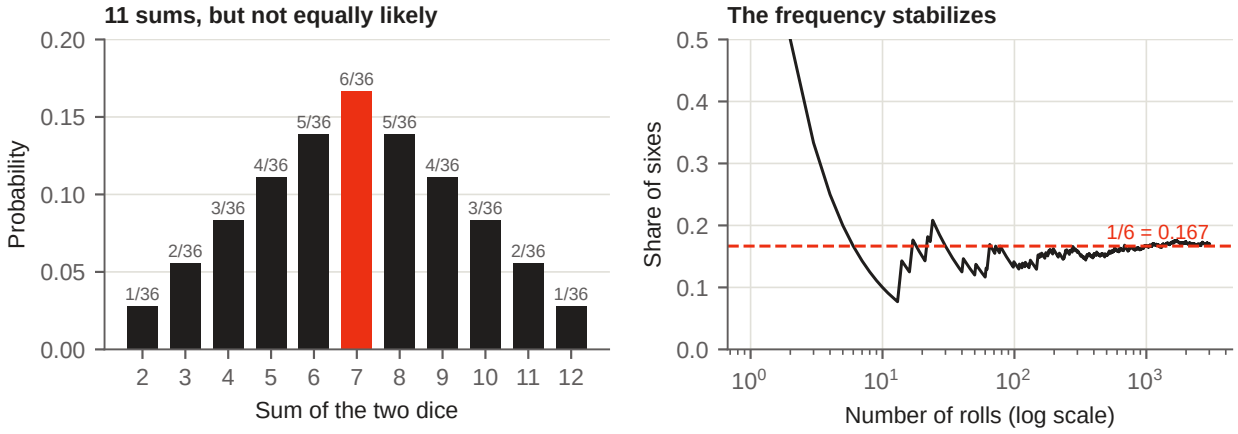


Fig. 1. Left: the sum of two dice, the 11 values are not equally likely. Right: the share of sixes in successive rolls of one die fluctuates at the start but over time approaches $1/6$.

3.1 Three sources of probability

- **Classical**: $P(A) =$ favourable/all; requires equally likely outcomes (dice, cards, a draw).
- **Frequentist**: $P(A) \approx$ number of occurrences/number of trials. Example: of 400 applications the bank approved 260, so $P(\text{approval}) \approx 0.65$. The right-hand panel of Fig. 1 shows why this works: the more trials, the closer the frequency to the "true" probability (the law of large numbers, week 7).
- **Subjective**: a weighted degree of belief ("a 70% probability of a rate cut"). It too must obey the axioms.

4 The complement and addition rules

COMPLEMENT RULE

$$P(A') = 1 - P(A). \tag{4.1}$$

Especially useful for "at least one" questions: it is easier to find the probability that *none occurs* and subtract it from 1.

EXAMPLE 2. CALL CENTRE

8% of calls are lost (the caller hangs up). Calls are independent; four came in. The probability that none is lost is $0.92^4 = 0.716$, so

$$P(\text{at least one lost}) = 1 - 0.92^4 = 1 - 0.716 = \mathbf{0.284}.$$

ADDITION RULE

$$P(A \cup B) = P(A) + P(B) - P(A \cap B). \tag{4.2}$$

The subtraction is needed so as not to count the intersection twice. For mutually exclusive events $P(A \cap B) = 0$ and what remains is $P(A \cup B) = P(A) + P(B)$.

EXAMPLE 3. BANK CUSTOMERS

Of 200 bank customers, 120 hold a debit card (D), 60 a credit card (C), 30 both. Then $P(D) = 0.60$, $P(C) = 0.30$, $P(D \cap C) = 0.15$ and

$$P(D \cup C) = 0.60 + 0.30 - 0.15 = \mathbf{0.75}, \qquad P(\text{no cards}) = 1 - 0.75 = \mathbf{0.25}\ (50 \text{ customers}).$$

If we had simply added $0.60 + 0.30 = 0.90$, the customers with two cards would have been counted twice.

COMMON MISTAKE

Probabilities may be added without subtracting the intersection only for mutually exclusive events. A typical mistake: "the default probability of each of three borrowers is 5%, so the probability that at least one goes bankrupt is 15%". This is wrong: the events are not mutually exclusive (several defaults may occur at once), and the sum overstates the answer.

5 Contingency table: joint, marginal and conditional probabilities

Data on two categorical attributes are conveniently written in a contingency table. Here are 400 loan applications: T — the applicant is from Tashkent, A — the application was approved.

	Approved (A)	Rejected (A')	Total
Tashkent (T)	168	72	240
Other regions (T')	92	68	160
Total	**260**	**140**	**400**

Three kinds of probabilities can be read from a single table:

- **Marginal** (from the total rows and columns): $P(A) = 260/400 = 0.65$; $P(T) = 240/400 = 0.60$.
- **Joint** (from the cells): $P(A \cap T) = 168/400 = 0.42$; $P(A' \cap T') = 68/400 = 0.17$.
- **Conditional** (within a row or column): $P(A \mid T) = 168/240 = 0.70$; $P(A \mid T') = 92/160 = 0.575$.

CONDITIONAL PROBABILITY

$P(A \mid B)$ is read "the probability of A given B" and is computed inside B:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B) > 0. \tag{5.1}$$

In the table: $P(A \mid T) = \dfrac{168/400}{240/400} = \dfrac{168}{240} = 0.70$. The denominator narrowed from 400 applications to the 240 from Tashkent. We shall study conditional probability in detail next week.

6 The multiplication rule and independence

Let us rewrite (5.1) in a form convenient for calculation:

$$P(A \cap B) = P(A) \cdot P(B \mid A). \tag{6.1}$$

"For both events to occur, the first must occur, and then the second, given that the first occurred."

INDEPENDENT EVENTS

Events A and B are **independent** if the occurrence of one does not change the probability of the other:

$$P(A \cap B) = P(A)\,P(B), \qquad \text{equivalently} \quad P(A \mid B) = P(A). \tag{6.2}$$

EXAMPLE 4. ARE REGION AND APPROVAL INDEPENDENT?

From the table $P(A) \cdot P(T) = 0.65 \times 0.60 = 0.39$, while in fact $P(A \cap T) = 0.42$. Since $0.42 \neq 0.39$, the events are *dependent*: approval is more likely for applicants from Tashkent (0.70 against the overall 0.65).

MUTUALLY EXCLUSIVE $\neq$ INDEPENDENT

Mutually exclusive events cannot occur together: $P(A \cap B) = 0$. This answers the question "is it possible for both at once?". **Independent** events carry no information about each other: $P(A \cap B) = P(A)P(B)$. This answers the question "does one change the probability of the other?". If $P(A) > 0$, $P(B) > 0$ and the events are mutually exclusive, they are *necessarily dependent*: having learned that A occurred, we know for certain that B did not.

For several independent events the probabilities are simply multiplied: $P(A_1 \cap \cdots \cap A_k) = P(A_1) \cdots P(A_k)$.

EXAMPLE 5. THREE BORROWERS

The default probability of each of three independent borrowers is 0.05. The probability that nobody defaults is $0.95^3 = 0.857$. Therefore

$$P(\text{at least one default}) = 1 - 0.95^3 = \mathbf{0.143} \quad (\text{and not } 0.15).$$

The naive sum $3 \times 0.05 = 0.15$ slightly overstates the answer: the probability of two and three simultaneous defaults has been counted an extra time. The more borrowers or the higher the probability, the larger the error.

THE GAMBLER'S FALLACY

A coin has come up heads five times in a row. The probability of heads on the sixth toss is still 0.5: the coin does not remember past tosses (tosses are independent). But the probability of six heads in a row, in advance, is $0.5^6 = 1/64 = 0.0156$. These are two different questions.

7 Combinatorics: how to count the number of ways

Formula (2.1) requires us to be able to count favourable and all outcomes. Combinatorics serves this purpose.

7.1 The multiplication principle

MULTIPLICATION PRINCIPLE

If the first choice can be made in n_1 ways, after it the second in n_2 ways, and so on, the total number of ways is $n_1 \cdot n_2 \cdots n_k$.

EXAMPLE 6

A menu of 3 starters, 4 main courses and 2 desserts gives $3 \cdot 4 \cdot 2 = 24$ different meals. A four-digit PIN (digits may repeat): $10^4 = 10\,000$ options; with distinct digits: $10 \cdot 9 \cdot 8 \cdot 7 = 5\,040$.

7.2 Factorial and permutations

Factorial: $n! = n \cdot (n-1) \cdots 2 \cdot 1$, by definition $0! = 1$. The number of ways to line up n distinct objects in a row is $n!$ (*permutations*). Five applications can be reviewed in $5! = 120$ different orders.

7.3 Arrangements and combinations

We choose r objects out of n distinct ones. Two questions: does the order of selection matter? If yes, these are *arrangements* (ordered selections); if not, *combinations*.

Arrangements and combinations

Arrangements (order matters): $$P(n,r) = \frac{n!}{(n-r)!},$$

Combinations (order does not matter): $$C(n,r) = \binom{n}{r} = \frac{n!}{r!\,(n-r)!}.$$

Relationships: $\binom{n}{r} = P(n,r)/r!$, and $\binom{n}{r} = \binom{n}{n-r}$. Russian-language textbooks use the notation A_n^r and C_n^r; English ones (Newbold, OpenStax) use $P(n,r)$ and $\binom{n}{r}$.

EXAMPLE 7. ARRANGEMENT OR COMBINATION?

From 10 analysts we choose a **president, vice-president and treasurer**: the posts are different, order matters, $P(10,3) = 10 \cdot 9 \cdot 8 = 720$. From 12 employees we choose a **committee of 4**: order does not matter, $\binom{12}{4} = \dfrac{12 \cdot 11 \cdot 10 \cdot 9}{4!} = 495$.

7.4 Permutations with repeated elements

If among n objects n_1 are of one kind, n_2 of another, etc., the number of distinguishable arrangements is $\dfrac{n!}{n_1!\,n_2!\cdots n_k!}$. The word “STATISTICS” (10 letters: S×3, T×3, A×1, I×2, C×1) can be formed in $\dfrac{10!}{3!\,3!\,1!\,2!\,1!} = \dfrac{3\,628\,800}{72} = 50\,400$ ways.

7.5 Probability through combinatorics

EXAMPLE 8. QUALITY CONTROL

A batch has 20 items, 3 of them defective. An inspector picks 4 items at random (without replacement). All $\binom{20}{4} = 4\,845$ sets are equally likely.

- No defectives: we choose all four from the 17 good ones, $\binom{17}{4} = 2\,380$, so $P = 2\,380/4\,845 = \mathbf{0.491}$.
- Exactly one defective: $\binom{3}{1}\binom{17}{3} = 3 \cdot 680 = 2\,040$, $P = 2\,040/4\,845 = \mathbf{0.421}$.
- At least one defective: $1 - 0.491 = \mathbf{0.509}$.

General formula: $P = \dfrac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$, where N is the batch size, K the number of defectives, n the sample size, k the number of defectives in the sample.

EXAMPLE 9. THE “6 OUT OF 45” LOTTERY

The winning combination is one of $\binom{45}{6} = 8\,145\,060$ equally likely sets. The probability of winning with one ticket is $1/8\,145\,060 \approx 1.2 \cdot 10^{-7}$.

EXAMPLE 10. THE BIRTHDAY PARADOX

What is the probability that in a group of n people at least two share a birthday? We count through the complement:

$$P = 1 - \frac{365 \cdot 364 \cdots (365 - n + 1)}{365^n}.$$

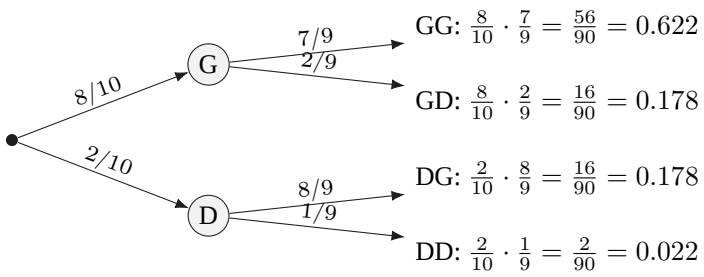
For $n = 23$ we get 0.507, for $n = 30$ we get 0.706. Already in a group of 23 a match is more likely than not, although intuition suggests much higher numbers.

8 The probability tree

When events occur in sequence, it is convenient to write the probabilities on a tree: along a branch the probabilities are *multiplied* (rule (6.1)), the probabilities of different leaves are *added*.

EXAMPLE 11. SELECTION WITHOUT REPLACEMENT

A batch has 10 items, 2 defective (D), 8 good (G). We take two items in a row without replacement.



Check: $56 + 16 + 16 + 2 = 90$, i.e. the leaf probabilities sum to 1. The probability of exactly one defective $= \frac{16}{90} + \frac{16}{90} = \frac{32}{90} = 0.356$. Combinatorics gives the same probability of DD: $\binom{2}{2}/\binom{10}{2} = 1/45 = 0.022$.

9 Key terms

Term	*In Russian*	Meaning
Random experiment	*Случайный эксперимент*	a procedure with an uncertain result
Sample space	*Пространство исходов*	the set of all possible outcomes, S
Event	*Событие*	a subset of S
Union	*Объединение*	$A \cup B$: A or B
Intersection	*Пересечение*	$A \cap B$: A and B together
Complement	*Дополнение*	A': not A
Mutually exclusive	*Несовместные*	cannot occur together
Independent	*Независимые*	$P(A \cap B) = P(A)P(B)$
Joint probability	*Совместная вероятность*	$P(A \cap B)$
Marginal probability	*Маргинальная вероятность*	$P(A)$, from the table totals
Conditional probability	*Условная вероятность*	$P(A \mid B)$, "A given B"
Factorial	*Факториал*	$n!$
Permutation (ordered selection)	*Размещение*	ordered choice of r from n
Combination	*Сочетание*	unordered choice of r from n

Formulas of Lecture 2

$$0 \le P(A) \le 1, \quad P(S) = 1, \quad P(A \cup B) = P(A) + P(B) \text{ (mutually exclusive)},$$

$$P(A') = 1 - P(A), \qquad P(A \cup B) = P(A) + P(B) - P(A \cap B),$$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \qquad P(A \cap B) = P(A)P(B \mid A),$$

$$\text{independence: } P(A \cap B) = P(A)P(B) \iff P(A \mid B) = P(A),$$

$$n!, \qquad P(n, r) = \frac{n!}{(n-r)!}, \qquad C(n, r) = \frac{n!}{r!(n-r)!}, \qquad \frac{n!}{n_1! \cdots n_k!}.$$

10 Practice problems

Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix "Statistical Tables".

Problem 1 [A] A questionnaire has two questions, each of which can be answered "yes" (Y) or "no" (N).

(a) Write down the sample space S.
(b) Write down the event "at least one answer is yes". How many outcomes does it contain?
(c) Find its probability, assuming all outcomes are equally likely.

Problem 2 [A]

(a) $P(\text{late}) = 0.30$. Find $P(\text{not late})$.
(b) $P(A) = 0.40$, $P(B) = 0.30$, $P(A \cap B) = 0.15$. Find $P(A \cup B)$.
(c) It is known that $P(A) = 0.20$ and $P(A \cap B) = 0.35$. What can you say about these numbers?
(d) Two dice are rolled. Find the probability that at least one six came up (in two ways).

Problem 3 [B] Of 500 firms, 300 export, 200 import and 120 do both. Find the probabilities that a randomly chosen firm: (a) exports or imports; (b) does neither; (c) only exports; (d) only imports.

Problem 4 [B] Of 300 students, 180 study Finance and 120 study Marketing. In the retake exam 90 finance students and 40 marketing students passed.

(a) Build the contingency table.
(b) Find $P(\text{passed})$, $P(\text{passed} \mid \text{finance})$, $P(\text{passed} \mid \text{marketing})$.
(c) Check whether the programme and passing are independent.

Problem 5 [B] An analyst claims: "the default probability of each of three borrowers is 0.05, so the probability that at least one of them goes bankrupt is $3 \times 0.05 = 0.15$". Point out two errors in this reasoning and find the correct value assuming the borrowers are independent.

Problem 6 [B] In how many ways can one:

(a) form a four-digit PIN code from distinct digits;
(b) choose a committee of 4 from 12 employees;
(c) choose a president, vice-president and treasurer from 10 analysts;
(d) rearrange the letters of the word "ECONOMICS"?

Problem 7 [B] A batch has 30 phones, 5 of them faulty. For inspection 3 are chosen at random (without replacement). Find the probabilities that among those inspected: (a) none is faulty; (b) exactly one is faulty; (c) at least one is faulty.

Problem 8 [B] (a) Find the probability that among 25 people there are at least two with a common birthday (assume the year has 365 days and all dates are equally likely). (b) Derive the formula and explain why it is easier to compute through the complement. (c) Find the answer for a group of 40 people.

Problem 9 [C] A bank issued 600 loans: 360 to employees, of whom 18 stopped paying, and 240 to entrepreneurs, of whom 36 stopped paying.

(a) Build the contingency table "borrower category $\times$ default".
(b) Find $P(\text{default})$, $P(\text{default} \mid \text{entrepreneur})$, $P(\text{default} \mid \text{employee})$.
(c) Are borrower category and default independent?
(d) Find $P(\text{entrepreneur} \mid \text{default})$ and explain how this probability differs from $P(\text{default} \mid \text{entrepreneur})$.

Problem 10 [C] For three events $P(A) = 0.50$, $P(B) = 0.40$, $P(C) = 0.30$, $P(A \cap B) = 0.20$, $P(A \cap C) = 0.15$, $P(B \cap C) = 0.10$, $P(A \cap B \cap C) = 0.05$.

(a) Find $P(A \cup B \cup C)$ by the inclusion–exclusion formula

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) \\ - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C).$$

(b) Find the probability that none of the three events occurs.

(c) Explain why the triple intersection is added at the end.

Problem 11 [C] It is known that $P(A) = 0.30$ and $P(B) = 0.50$.

(a) Let A and B be mutually exclusive. Find $P(A \cup B)$. Can they be independent?

(b) Let A and B be independent. Find $P(A \cap B)$ and $P(A \cup B)$.

(c) Can events be mutually exclusive and independent at the same time?

Problem 12 [C] In the 17th century the Chevalier de Méré considered two bets equivalent. (a) What is the probability of rolling at least one six in 4 rolls of one die? (b) What is the probability of at least one "double six" in 24 rolls of two dice? Which bet is more favourable? (c) A coin has landed heads 5 times in a row. What is the probability of heads on the next toss, and why is it not > 0.5?